



OPERATING STATISTICS

1. INTRODUCTION

Operating Statistics are essential for planning, prioritizing and monitoring railway operations. They help in evaluating **performance**, identifying **bottlenecks**, and optimizing **traffic movement**.

Railway statistics are based on **four key parameters**:

Quantity • Distance • Duration • Service

2. PRIMARY UNITS (Basic Statistical Units) are basic units for measuring

Primary units measure fundamental quantities directly associated with railway operations.

(a) Quantity (weight)

Expressed as:

- **Tonnes** (Goods carried)
- **Number of Passengers** carried
- **Earnings** derived (Passenger / Goods earnings)

(b) Distance

Measured in:

- **Kilometres (Km)**

(c) Time (Duration)

Measured in:

- **Minutes**
- **Hours**
- **Days**

(d) Service Performed

Expressed in terms of:

- **Trains**
- **Vehicles**
- **Wagons**
- **Engines (Locomotives)**

✂ Primary units are the building blocks of all operating statistics.

3. FUNDAMENTAL UNITS (Composite Units)

These are relationships between primary units and express **two primary ideas combined**.

★ Examples of Fundamental Units

Fundamental Unit Relation		Meaning
Tonne-km	Tonnes × Km	Goods traffic carried over distance
Passenger-km	Passengers × Km	Passenger traffic carried over distance
Train-km	Train run × Km	Total distance covered by trains
Wagon-km	Wagons × Km	Wagon movement over distance
Engine hours	Time Loco in use	Utilisation of locomotive
Wagon-days	Time wagons in holding	Availability measure

✂ Fundamental units reflect actual transportation work done.

4. DERIVED UNITS (Performance Indicators)

Derived units express the **relationship between two sets of primary or fundamental units**.

They highlight:

- Productivity
- Efficiency
- Operational performance

Examples (Very Important for MCQs & Viva)

(a) Passenger Earnings ÷ Passengers Carried = Earnings per Passenger

- Measures revenue efficiency per passenger.

(b) Passenger Earnings ÷ Passenger-km = Earnings per Passenger-km

- Measures revenue per unit traffic work.

(c) Passenger-km ÷ Number of Passengers = Average Lead

- Lead of Passenger Traffic
- Indicates average distance travelled per passenger.

 Frequently asked in AOM Viva.

(d) $\text{Wagon-km} \div \text{Wagon-days} = \text{Wagon-km per Wagon-day}$

- Key indicator of wagon utilisation.

5. PURPOSE & IMPORTANCE OF OPERATING STATISTICS

Operating statistics help in:

- Traffic planning
- Capacity utilization monitoring
- Improving wagon/loco utilization
- Evaluating throughput
- Decision making for asset deployment
- Controlling expenditure
- Developing operating strategies

They support both **strategic** and **tactical** railway management.